

Muzammil Nawaz Khan

Computer Vision Engineer

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SUMMARY

Computer vision and medical imaging engineer who benchmarked 3D U-Net against SegFormer3D for multi-modal MRI brain tumor segmentation on BraTS20 (85%+ Dice), added GradCAM explainability and served the model with FastAPI. The work won 1st Place at COMPPEC 2025 and a Departmental Silver Medal. Also 1+ year of production ML experience in Python (PyTorch, TensorFlow, OpenCV). Open to remote work and relocation.

SKILLS

Computer Vision: OpenCV, U-Net / 3D U-Net, SegFormer / SegFormer3D, YOLO, GradCAM, medical image segmentation
Machine Learning: Python, PyTorch, TensorFlow, Keras, scikit-learn, CatBoost, XGBoost, AutoGluon, Hugging Face Transformers, Pandas, NumPy
Backend & Infrastructure: FastAPI, Django / DRF, REST APIs, Docker, Kubernetes, Git/GitHub, CI/CD
Generative AI & Agents: OpenAI API, Anthropic API, LLMs, prompt engineering, structured outputs, RAG, LLM-as-judge evaluation, LangChain, LangGraph, CrewAI, AutoGen, Model Context Protocol (MCP), n8n, ChromaDB, FAISS, custom HNSW, BM25, Reciprocal Rank Fusion
Languages: Python, SQL, Rust, C++, JavaScript/TypeScript
Domain: Healthcare IT, medical imaging, AI training data

EXPERIENCE

AI/ML Research Intern, Risetech Jul 2024 – Aug 2024
Rawalpindi / Islamabad, Pakistan

- Segmented brain tumor regions from multi-modal MRI (T1-CE, T2, FLAIR) on the BraTS20 dataset.
- Benchmarked 3D U-Net against SegFormer3D and selected SegFormer3D (85%+ Dice). Used GradCAM for explainability and FastAPI for serving.
- The work became the Neurofusion final year project: 1st Place at COMPPEC 2025 and a Departmental Silver Medal.

Junior AI Engineer, CareCloud (MTBC) Jul 2025 – Aug 2026
Islamabad, Pakistan

- Delivered a production claims-classification system (CatBoost, XGBoost, AutoGluon) at 95% parent-model accuracy and 92%+ category-level accuracy across 42,900+ cases, with daily feedback-loop retraining and multi-month tracking of accuracy and stability.
- Engineered Python pipelines that pull Hikvision NVR logs, clean them and load them to a database for HR attendance analytics; built an internal FastAPI EDI Viewer for inspecting raw healthcare claim data.
- Built a Python denial-prediction and resubmission automation system with a rule-based First-Time Pass Rate (FTPR) module comparing rejected vs corrected/paid claims by provider, payer, CPT and ICD code. Cut resubmission wait from 2 weeks to 3 days and reduced the denial rate from 3.2% to 2.4%.

Software / AI Engineer, Suidhaga Mar 2022 – Nov 2025
Remote, part-time alongside studies

- Applied data analysis and data science to business data; later built ARIMA-family forecasting models to predict product prices.

Freelance AI Engineer, Independent Sep 2022 – Present
Remote

- Delivered AI development projects for clients, including model training and applied AI application builds.

PROJECTS

Neurofusion: 3D Brain Tumor Segmentation (PyTorch, SegFormer3D, 3D U-Net, GradCAM, FastAPI): Final year project on multi-modal MRI (T1-CE, T2, FLAIR) from the BraTS20 dataset. SegFormer3D selected over 3D U-Net at 85%+ Dice, with GradCAM explainability. 1st Place at COMPPEC 2025 and a Departmental Silver Medal.

Computer Vision Suite (Python, OpenCV): Retinal vessel segmentation (multi-level thresholding, morphology), skin lesion segmentation (connected component labeling, IoU), optical Braille recognition (contour analysis), license plate localization (edge detection), optic disc localisation.

Luggage Threat Detection (Python, ANN, OpenCV): ANN architecture for image classification of potential threats in luggage images.

RAG Custom Engine (Pure Python, GPT-4o-mini): From-scratch hybrid retrieval with a custom HNSW store, Okapi BM25 and Reciprocal Rank Fusion, plus a 3-stage Self-RAG gate (retrieval decision, relevance grading, hallucination check) and contextual compression. About 20% precision gain over dense-vector baselines.

EDUCATION

Bachelor of Computer Engineering, National University of Sciences and Technology (NUST) Nov 2021 – Jul 2025
Islamabad, Pakistan
CGPA 3.25. Final year project: Neurofusion, 3D brain tumor segmentation (SegFormer3D, BraTS20, 85%+ Dice); 1st Place COMPPEC 2025, Departmental Silver Medal.

F.Sc. Pre-Engineering, Ghazali Premier College Jun 2018 – Aug 2020
Lahore, Pakistan

CERTIFICATIONS

- IBM Deep Learning with PyTorch, Keras and TensorFlow Professional Certificate
- Informed Clinical Decision Making using Deep Learning, University of Glasgow
- IBM AI Engineering Professional Certificate
- Generative AI Engineering with LLMs Specialization (IBM)
- Large Language Model Operations (LLMOps) Specialization, Duke University

HONORS & AWARDS

1st Place, COMPPEC 2025 | Departmental Silver Medal